OEFF Prediction Model

Ultimate Frisbee Offensive Efficiency Analysis

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Dataset Rows	Features Used	Target Variable	CV Folds	Model Type
1748	7	OEFF	5	RandomForest

1. Executive Summary

This report presents the results of a comprehensive machine learning analysis to predict team Offensive Efficiency (OEFF) in Ultimate Frisbee based on individual player statistics. The analysis evaluated multiple regression models and feature combinations using 5-fold cross-validation on data from 1748 player-season records.

Key Findings:
• Best Model: RandomForest
• Cross-Validated R^2: 0.4247
• In-Sample R^2: 0.9223
• Features Used: 7 out of 13 candidates
• RMSE: 2.2423

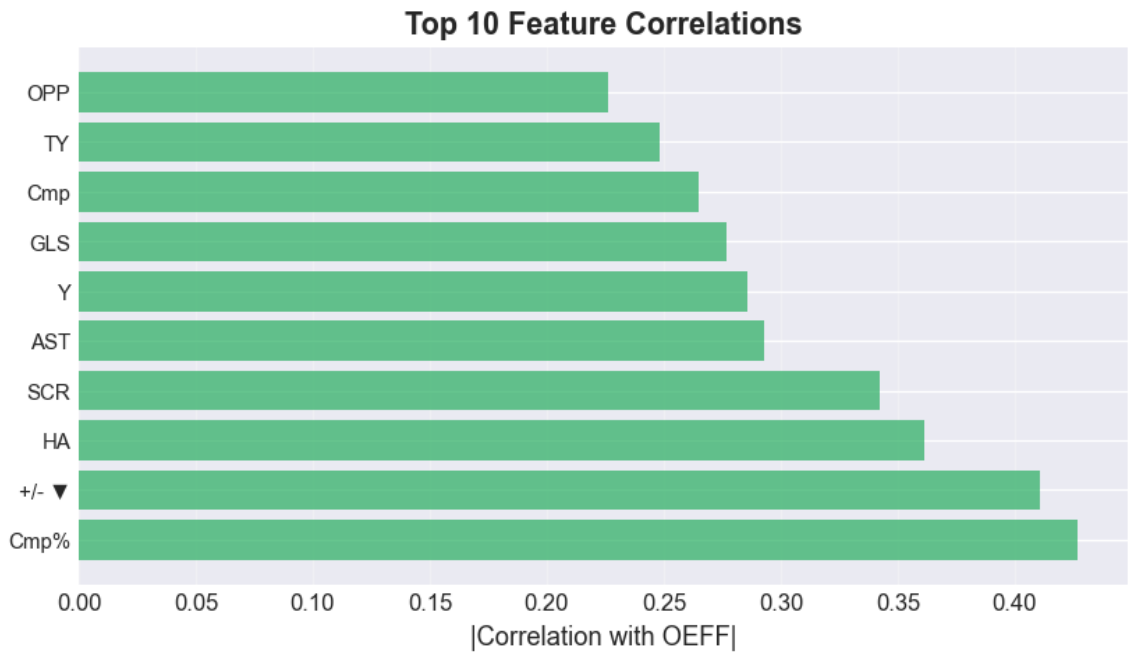
Selected Features: +/- ▼, Cmp%, OPP, TY, G, PP, HA

2. Feature Correlation Analysis

Features were filtered based on their correlation with OEFF. Correlation threshold: |r| ≥ 0.2

13 of 24 candidate features passed the threshold.

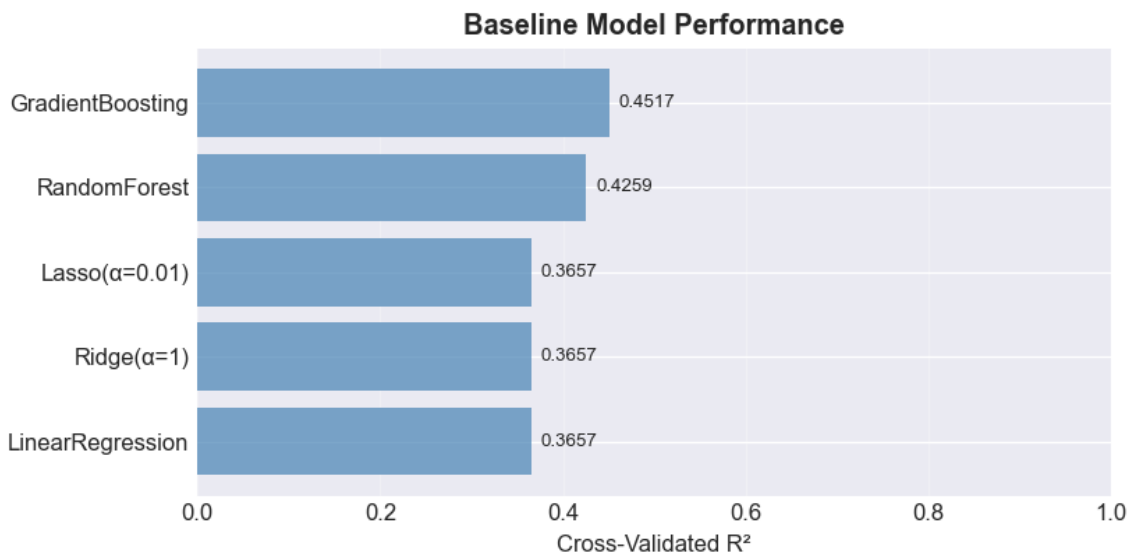
Feature	Correlation	Included
Cmp%	0.4269	Yes
+/- ▼	0.4106	Yes
HA	0.3616	Yes
SCR	0.3421	Yes
AST	0.2926	Yes
Y	0.2856	Yes
GLS	0.2769	Yes
Cmp	0.2651	Yes
TY	0.2483	Yes
OPP	0.2260	Yes
RY	0.2194	Yes
PP	0.2165	Yes
G	0.2129	Yes
MP	0.1951	No
Hck%	0.1745	No



3. Baseline Model Comparison

Five regression models were evaluated using all 13 filtered features to establish baseline performance.

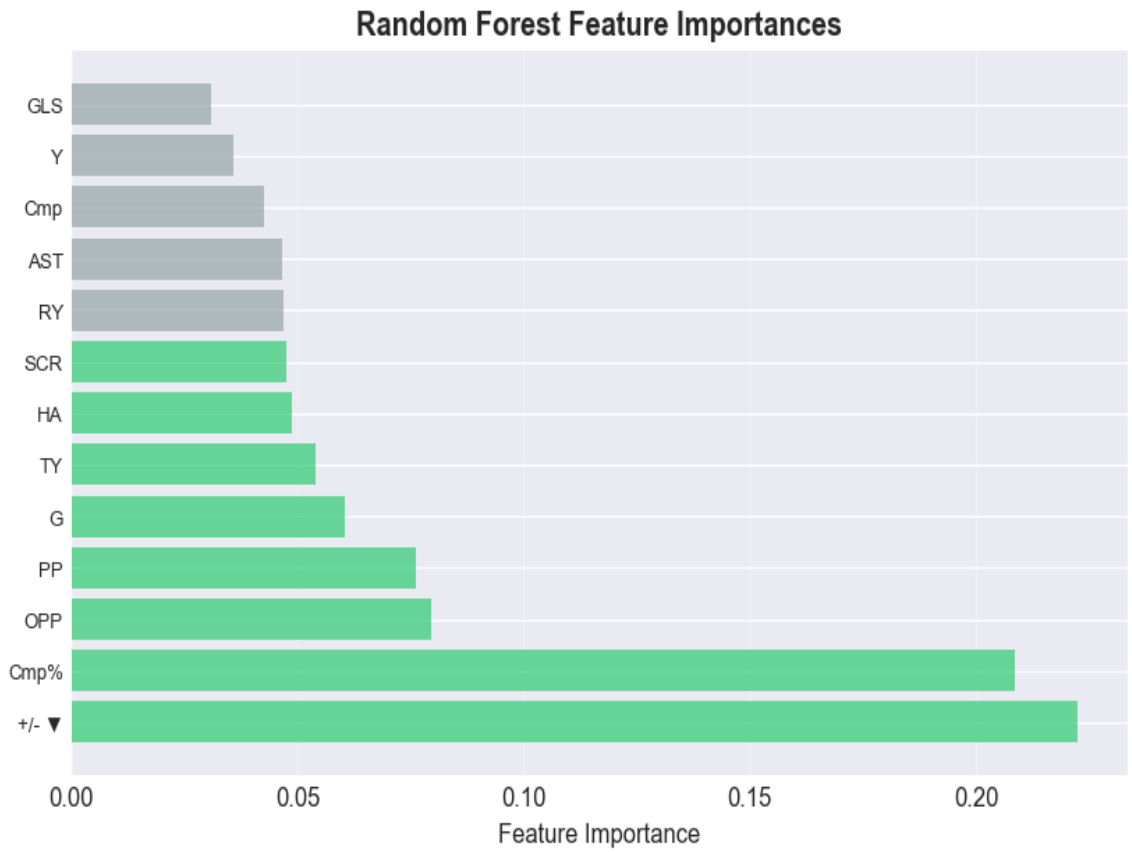
Model	Features	CV R^2	CV RMSE
GradientBoosting	13	0.4517	5.9496
RandomForest	13	0.4259	6.0861
LinearRegression	13	0.3657	6.4010
Ridge(α=1)	13	0.3657	6.4007
Lasso(α=0.01)	13	0.3657	6.4007



4. Feature Importance Rankings

Random Forest feature importance analysis identified the most predictive features. The top 8 features were selected for exhaustive subset search.

Rank	Feature	Importance	In Subset Search
1	+/- ▼	0.2224	✓
2	Cmp%	0.2087	✓
3	OPP	0.0794	✓
4	PP	0.0763	✓
5	G	0.0603	✓
6	TY	0.0540	✓
7	HA	0.0486	✓
8	SCR	0.0474	✓
9	RY	0.0470	
10	AST	0.0467	
11	Cmp	0.0425	
12	Y	0.0358	
13	GLS	0.0308	


Random Forest Feature Importances
GLS
Y
Cmp
AST
RY
SCR
HA
TY
G
PP
OPP
Cmp%
+/- ▼
0.00
0.05
0.10
0.15
0.20
Feature Importance

5. Feature Subset Selection Results

Forward selection was performed on the top 8 features using three model types. The best combination achieved a cross-validated R^2 of 0.4247.

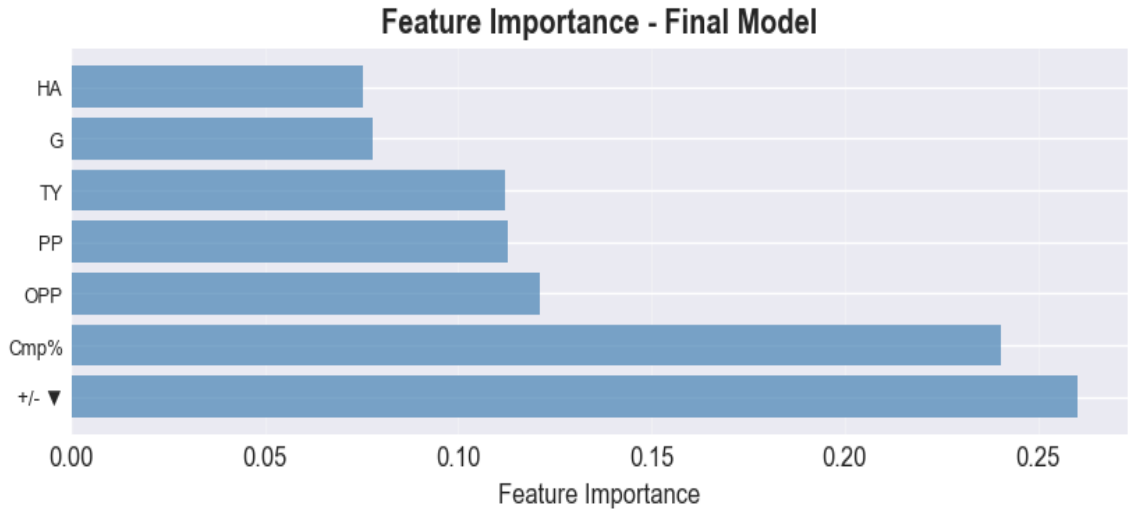
Rank	Model	Features	CV R^2	CV RMSE
1	RandomForest	+/- ▼, Cmp%, OPP...	0.4247	6.0931
2	RandomForest	+/- ▼, Cmp%, OPP...	0.4237	6.0961
3	RandomForest	+/- ▼, Cmp%, OPP...	0.4140	6.1477
4	RandomForest	+/- ▼, Cmp%, OPP...	0.4120	6.1573
5	RandomForest	+/- ▼, Cmp%, OPP...	0.4103	6.1644
6	RandomForest	+/- ▼, Cmp%, OPP...	0.4010	6.2112
7	RandomForest	+/- ▼, Cmp%, OPP...	0.3958	6.2374
8	RandomForest	+/- ▼, Cmp%, OPP...	0.3844	6.3013
9	RandomForest	+/- ▼, Cmp%, OPP...	0.3672	6.3839
10	RandomForest	+/- ▼, Cmp%, OPP...	0.3613	6.4094

6. Final Model Performance

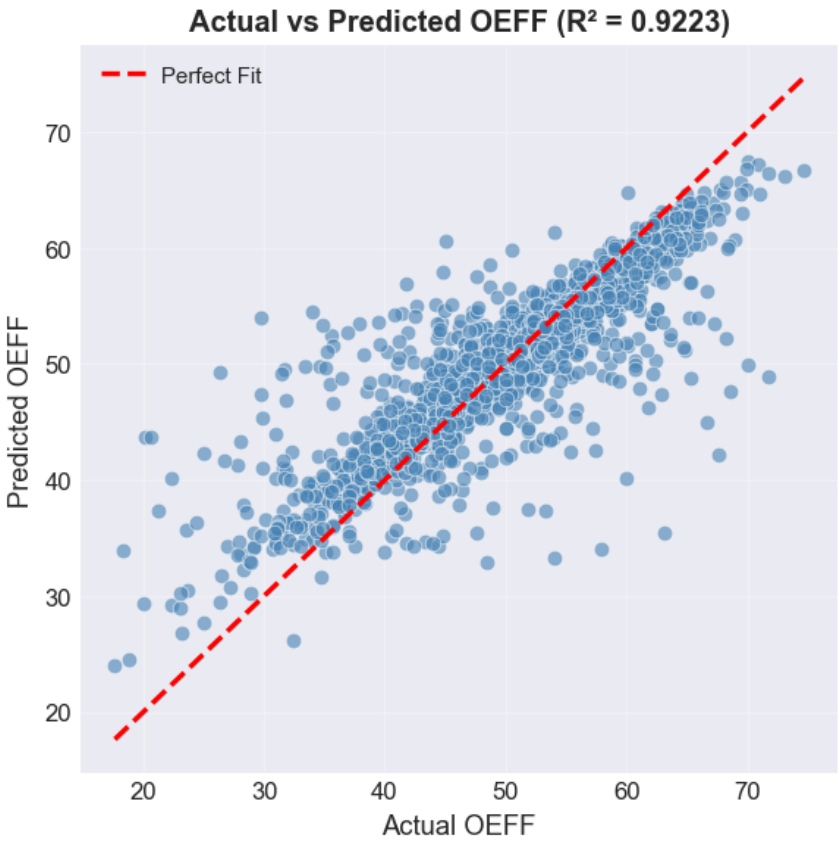
Metric	Value
Model Type	RandomForest
Number of Features	7
Cross-Validated R^2	0.4247
In-Sample R^2	0.9223
In-Sample RMSE	2.2423

6.1 Feature Importance

Feature	Feature Importance
+/- ▼	0.2602
Cmp%	0.2403
OPP	0.1211
PP	0.1128
TY	0.1121
G	0.0779
HA	0.0755



7. Model Validation



Residual Analysis:
Mean Absolute Error: 2.6585
Mean Percent Error: 5.88%
Std of Residuals: 4.2434

8. Player-Level Predictions

The final model was applied to predict OEFF for all players in the dataset. Predictions were successfully generated for 2149 players.

8.1 Top Over-Performers (Actual > Predicted)

Player	Team	Actual	Predicted	Difference
Thomas Hansen	thunderbirds	63.16	35.45	+27.71
Adrian Golay	alleycats	67.65	42.17	+25.48
Collin Gall	growlers	57.89	34.11	+23.78
Alex Thomas	thunderbirds	71.74	48.86	+22.88
Jac Carreiro	glory	66.67	45.03	+21.64

8.2 Top Under-Performers (Actual < Predicted)

Player	Team	Actual	Predicted	Difference
Patrick Xu	spiders	29.69	53.96	-24.27
Colin Ringwood	mechanix	20.19	43.72	-23.53
Terry Gaither	mechanix	20.69	43.72	-23.03
Avi Taylor	steel	26.32	49.24	-22.92
Cal Tornabene	union	34.02	54.56	-20.54

9. Conclusions and Recommendations

Key Findings:
1. The RandomForest achieved $R^2 = 0.4247$, explaining 42.5% of variance in team OEFF.
2. Key predictive features: +/- ▼, Cmp%, OPP and 4 others.
3. Low prediction error (RMSE = 2.2423).

Applications: Player evaluation, roster construction, performance tracking, draft decisions.

Next Steps: Validate on future seasons, explore position-specific models, investigate feature interactions.